



## ya diggg: 'All-American' Manufacturing & Sourcing Strategy

Pivoting the ya diggg to an **All-American made and assembled** model elevates the brand to "Heirloom Hardware" status. This strategy focuses on domestic high-precision manufacturing assembly, and compliance with FTC "Made in USA" standards

### 1. The USA Manufacturing Tier

To maintain the premium "Teenage Engineering/ Leica" quality within the US, we have identified specialized boutique contract manufacturers (EMS) and fabrication houses.

| Service                 | Recommended USA Partner | Location        | Capability Rationale                                                                                                 |
|-------------------------|-------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------|
| Turnkey Assembly        | Sierra Circuits         | San Jose, CA    | Leader in high-precision, ITAR-certified US assembly and NPI (New Product Introduction) <a href="#">Protoexpress</a> |
| Enclosure (CNC/Molding) | Cypress Industries      | Austin, TX      | Specializes in high-end plastic injection molding and CNC aluminum tooling <a href="#">Cypressindustries</a>         |
| Enclosure (Boutique)    | Protolabs               | Maple Plain, MN | Rapid US-based CNC for aluminum frames and translucent polycarbonate <a href="#">Youtube</a>                         |
| PCB Fabrication         | Advanced Circuits       | Aurora, CO      | One of the largest US-based PCB fab houses, 100% domestic capability <a href="#">Fscircuits</a>                      |

### 2. Updated 'Made in USA' BOM Estimate (Low MOQ: 500 Units)

Manufacturing in the USA carries a **1.6x - 2.2x labor/overhead premium** compared to global sourcing. However, for a high-margin niche device, this is offset by lower tariffs, reduced logistics risk, and premium branding.

| Category              | Global Target | USA Estimate | US Impact Note                                                             |
|-----------------------|---------------|--------------|----------------------------------------------------------------------------|
| Core ICs (Global)     | \$42.00       | \$42.00      | Chips (NXP, Micron) are global commodities.                                |
| Imaging & AF          | \$18.50       | \$21.00      | Higher logistics/buffer for US-based sensor distributors                   |
| Display Unit          | \$14.00       | \$16.00      | Sourcing from US-based agents for Sharp.                                   |
| Mechanicals           | \$22.00       | \$48.00      | Domestic CNC/Molding is significantly higher cost <a href="#">Facebook</a> |
| PCB & Passive         | \$12.00       | \$24.00      | US-based quick-turn fab and passive sourcing                               |
| Assembly & Test       | \$4.00 (Est)  | \$18.50      | Living-wage US technical labor for precision assembly                      |
| Packaging & Logistics | \$4.00        | \$8.00       | Premium US-made sustainable packaging.                                     |
| TOTAL BOM             | ~\$119.00     | ~\$177.50    | <b>+49% Cost Increase</b>                                                  |

### 3. The "Made in USA" Sourcing Reality

To legally claim **"Made in USA"** under FTC guidelines, the product must be **"all or virtually all"** made in the United States.

- **Silicon Reality:** No commercial mobile-grade SoCs (NXP) or high-end image sensors are currently manufactured in the US at this scale.
- **Compliance Strategy:** The FTC allows for imported "unavoidable" global components (ICs, Displays) if the **substantial transformation** (Assembly, PCBA, Final Testing) and the **high-value mechanicals** (CNC frame, molded shell) are domestic.
- **Proposed Label:** *"Designed and Assembled in [City], USA using American and Global Components"*

#### 4. Brand Positioning: "The American Tool"

Transitioning to US manufacturing transforms the ya diggg from a "consumer gadget" into a **"professional instrument."**

1. **The "Dungeon Family" Connection:** Lean into the Atlanta/Southern hip-hop roots by sourcing the final assembly in the **American South** (e.g., Texas or Georgia CMs) to anchor the story.
2. **Repairability:** US assembly allows for a "Mail-in Repair" service, aligning with the **Leica/Braun** heirloom philosophy of a product that lasts for decades.
3. **Transparency:** The translucent shell now reveals an **"Assembled in USA"** silk-screened on the internal aluminum frame, visible through the smoked amber plastic.

#### 5. Next Execution Steps

- ☐ **Request RFQs:** Send the finalized BOM to **Sierra Circuits** and **Cypress Industries** for firm 500-unit US production quotes.
- ☐ **DFM for USA CNC:** Adjust the internal aluminum frame design to optimize for US-based CNC machine rates (\$75-\$120/hr).
- ☐ **FTC Audit:** Consult a trade specialist to ensure the "substantial transformation" occurs domestically to protect the "Made in America" marketing claim.